

Accurate settlement data supports disciplined investigation

GB Settlement & Imbalance Service Intelligence Monitor

Elexon data	Validation	Detection	Investigation	Next action
Preserve the original evidence	Check coverage, values and time	Screen unusual periods	Separate known facts from hypotheses	Recommend evidence checks before escalation

Independent portfolio project. Not affiliated with or endorsed by Elexon.

The workflow preserves evidence at each analytical stage

Official API	Immutable raw	Validated data	Semantic model	Reporting
Daily prices and selected price stacks	Original bytes, request receipts and SHA-256 hashes	Python controls and independent SQL summaries	Date and period dimensions Explicit measures	Four report pages Structured investigation records

OBSERVED Source values

DERIVED Calculations

FLAG Review rules

HYPOTHESIS Explanations still to test

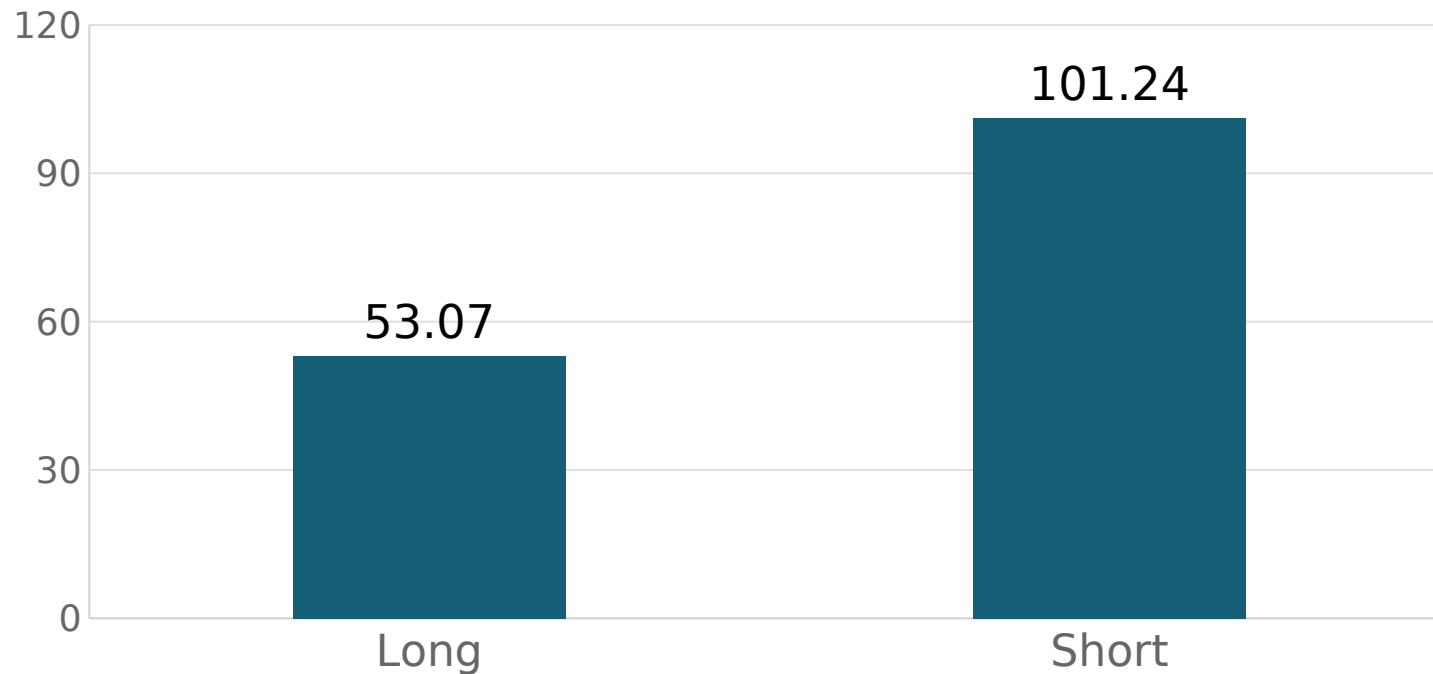
All 4,418 expected periods pass the core data checks

Control	Snapshot result	Interpretation
Coverage	4,418 / 4,418	100% of expected periods
Duplicate keys / core nulls	0 / 0	No detected core defects
Schema / invalid values	0 core exceptions	Required fields pass
Clock change	50 periods on 26 Oct	London calendar check passes
Retrieval	92 / 92 daily requests	Successful snapshot retrieval

Source creation time and retrieval time remain separate. Later source revisions are possible.

Short-system periods had higher prices in this quarter

Mean System Price (GBP/MWh)



2,340 long periods
2,076 short periods

Median prices
Long: £62.28/MWh
Short: £98.73/MWh

Two balanced periods sit outside this comparison.

130 periods meet the retrospective review thresholds

92 price flags + 45 imbalance flags – 7 overlapping periods = 130 distinct periods

Selected case	System Price	NIV	Reason
2025-10-13 SP26	£487.00/MWh	+184.92 MWh	Maximum price
2025-11-01 SP19	£-30.00/MWh	-1,136.75 MWh	Minimum price
2025-10-03 SP37	£109.40/MWh	+1,563.34 MWh	Largest NIV

Price $\leq -£11.22$ or $\geq £153.8997/\text{MWh}$. Absolute NIV ≥ 876.0417 MWh.

Three cases have detailed price evidence. The other 127 remain at detection level.

The £487 price reconciles, while the underlying cause remains open

KNOWN

13 October 2025, SP26
System Price: £487/MWh
NIV: +184.92 MWh (short)
Core quality checks pass

Retained adjusted cost / volume
£479.6670462 / 0.9849426 MWh
+ £0/MWh adjuster = £487/MWh

NOT ESTABLISHED

The cause of the short position
Why particular actions were needed
Any platform or service incident

The stack supports the price calculation.
It does not identify the market cause.
Matching prices do not prove identical
source vintages.

Further checks should test explanations before escalation

Evidence stage	Current position	Next check
Core data	Passed for this snapshot	Recheck revisions before reuse
Price calculation	Three cases reconciled	Compare calculation vintages
Balancing context	Partial action evidence	Inspect action timing and adjustments
Demand / generation	Cause still open	Test a specific supply/demand hypothesis
Service incident	No attribution established	Check official status and issue evidence

Escalation needs an evidence-backed issue, a clear owner and a documented next action.

The project makes Service Analyst work visible and reviewable

Capability	Project evidence
Analysis	Python / SQL cross-checks and descriptive statistics
Validation	Executable controls, clock-change tests and source receipts
Investigation	130-case queue and three price reconciliations
Communication	Known / unknown / next-check records and insight note
Process improvement	Repeatable ingestion and validated rebuild workflow

Power BI report definitions pass structural/schema checks. Live model calculations, screenshots and interactions still require a licensed runtime.